

Related Roadmaps

✓

 DevOps Roadmap

✓

 System Design Roadmap

✓

 Shell/Bash Roadmap

✓

 Cybersecurity Roadmap

Basic Terminology

Client

Server

Host

Package

Frame

Bandwidth

Latency

Throughput

Protocol

Port

Socket

IP Address

Mac Address

ARP

Transmission Media Types

Core Protocols

HTTP / HTTPS

SSH

SSL / TLS

NTP

FTP / SFTP

SNTP

SMTP / IMAP

DHCP

Network Access

Internet

Transport

Application

IPv4 vs IPv6

Public vs Private Addresses

IP vs MAC vs ARP

Static vs Dynamic Routing

Default Gateway

VLANs

STP

Link Aggregation

MAC Address Tables

WiFi Standards

Access Points & Controllers

Bluetooth Basics

Hotspot and Tethering

Mobile Networks

Wireless Security

WPA vs WPS

IPSec vs SSL VPN

Site-to-Site vs Remote Access

DoS & DDoS

Network Observability Tools

Wireshark

Nmap

NetFlow / sFlow

SNMP

Network Engineer

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

Introduction

How does the Internet Work?

What are Networks?

Network Types

LAN

WAN

MAN

WLAN

PAN

SAN

VPN

Cloud

Client-Server Network

Peer-to-Peer Network

Network Devices

Routers

Switches

Hub

Modems

Access Points

OSI Model

7 Layers

Physical

Data Link

Network

Transport

Session

Presentation

Application

TCP/IP Model

IP Addressing

Subnetting

Subnet Masks

CIDR

VLSM

Supernetting

Routing

Routing Protocols

BGP

OSPF

RIP

EIGRP

MPLS

Switching

DNS

DNS Servers

Cloudflare

Google

OpenDNS

Quad9

Wireless Networking

Load Balancer

Round Robin

Least Connections

Failover

QoS (Quality of Service)

Traffic shaping

Packet prioritization

Network Security

Cybersecurity Roadmap

Firewalls

VPNs

Encryption Basics

Network Attacks

IDS / IPS

Zero Trust Architecture

Firewall Types

Packet Filtering

Stateful Inspection

Next-Generation

Proxy

Circuit Level Gateway

Web Application

Observability

Visit the following roadmaps to keep learning

DevOps

Backend

System Design